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For this assignment, I implemented a Linux Kernel Module “kmlab” to track CPU time for registered processes. The module works with the “/proc” filesystem, more specifically through “/proc/kmlab/status”, allowing users to see a list of processes and their CPU times.

I created a "/proc/kmlab/status" entry to display process information and used a linked list to store "pid" and "cpu_time" for each registered process. During initialization ("kmlab_init"), I used "proc_mkdir" to create a "/proc" directory called "kmlab" and "proc_create" to add a status file. The "procfile_write" function handles writing to "/proc/kmlab/status" by adding the provided process ID to the linked list, while the "procfile_read" function outputs the PID and CPU time for all tracked processes. A timer callback ("my_timer_callback") is used to periodically update the CPU usage time for each tracked process.

I also developed a user-space application ("userapp") that writes process IDs to "/proc/kmlab/status". The application uses "getpid()" to get the current process ID and registers it with the kernel module.

After building and loading the kernel module with "insmod", I used "./userapp" to register a process, then checked "/proc/kmlab/status" to verify the output. The output showed the PID and CPU time as expected. Running "userapp" twice displayed both processes with their CPU runtimes, confirming multiple processes could be tracked simultaneously.

The implementation provided a simple way to track and display process CPU times through "/proc". The main challenge was ensuring smooth interactions between the linked list and "/proc", which was resolved after debugging.

Output from one process case

```
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ sudo insmod kmlab.ko
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ ./userapp
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ cat /proc/kmlab/status
PID      CPU_TIME
16944    10324000000
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$
```

Output from two process case

```
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ ./userapp &
[1] 17015
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ ./userapp
[1]+  Done                    ./userapp
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$ cat /proc/kmlab/status
PID      CPU_TIME
17045    6936000000
17015    7336000000
16944    10324000000
ubuntu@ubuntu:~/Desktop/360/pa-3-danielgapper12$
```